

CS319-1 Deliverable 2

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1- Non-Function Requirements

1.1 Usability

- **Notification System:** Our app is designed with a notification system that allows users to get informed when important events occur. For instance, an advisor will be notified when a high school applies for a tour if that tour is on the advisor's day of responsibility. Similarly, the guides will be notified when a tour is accepted and needs guides.
- **Automated Email System:** After certain events, actors will be notified by email. Such as, after the acceptance of the tour, high-school counselors will be notified. Also, new account information will be delivered to the student's Bilkent email when the coordinator creates a new account for the guide.
- **Dashboard System:** A dashboard for all actors of the Information Office to view all actions related to event planning in a user-friendly manner including the name of the closest high school visit, the number of upcoming tours and fairs.
- **Accessing the Tour Summary:** The key attributes, such as the date of the tour, number of students, and status of the tour, will be accessed by at most two button clicks.

1.2 Security/Safety

- **Account Creation:** The administrative accounts will be auto-generated by the system, and the credentials will be delivered by email. Furthermore, only the coordinator will be able to create accounts for new guides to prevent unauthorized access.
- **Limited Scope of Visible Information:** The information accessible to each user will be determined by their role in the system. For example, guides will only see essential details about system actors, such as their names, departments, and roles. More sensitive information, like email addresses and student numbers, will be restricted to the coordinator and Head to ensure privacy protection.
- **The credentials of the actors will be encrypted using an encryption algorithm and saved to the database.**

1.3 Performance

- After a guide assigns their self to a high-school tour, this change must be reflected in the system in less than 10 seconds to prevent collisions.
- All the tour interactions, such as canceling, rejecting, and accepting a tour, will be reflected in the system in less than 10 seconds.
- Page navigation within the app will not exceed 6 seconds.

1.4 Reliability

- The database structure will be organized and include backups that prevent data loss in the system.
- The system must accommodate more than 100 concurrent users to guarantee seamless access.

1.5 Maintainability

- MVC (Model-View-Controller) architecture will be implemented to separate the system's components clearly to clarify that each part can function independently. This modular structure will facilitate easier maintenance, testing, and scaling of individual system components.
- The microservice architecture will be implemented to enhance maintainability by allowing each service to be updated and maintained independently without affecting other parts of the system. This isolation will simplify debugging and testing. Moreover, it will enable us to work on different services simultaneously, increasing development efficiency, making version control process easier.

2- Activity Diagram

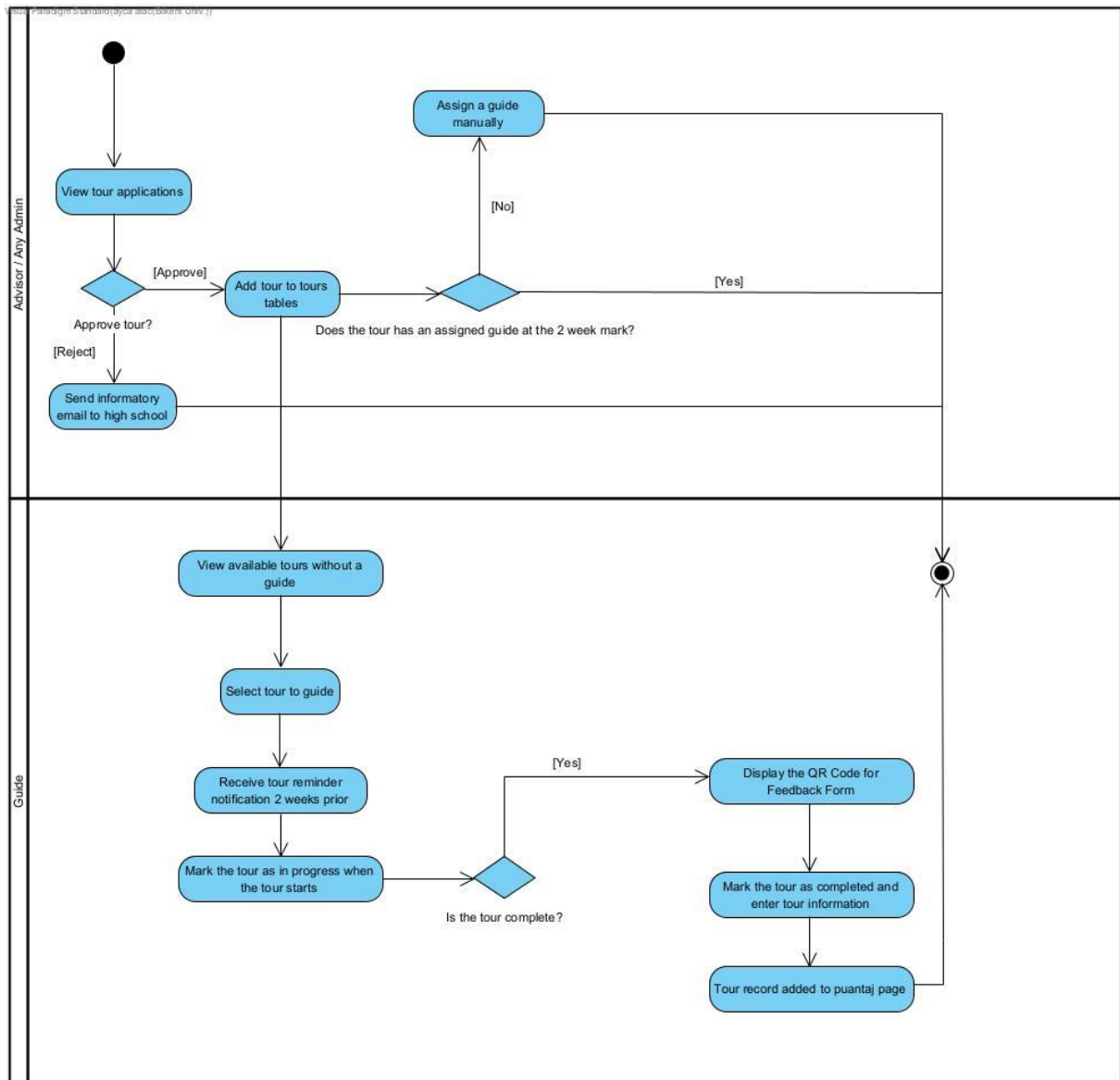


Figure 1: Tour Management System

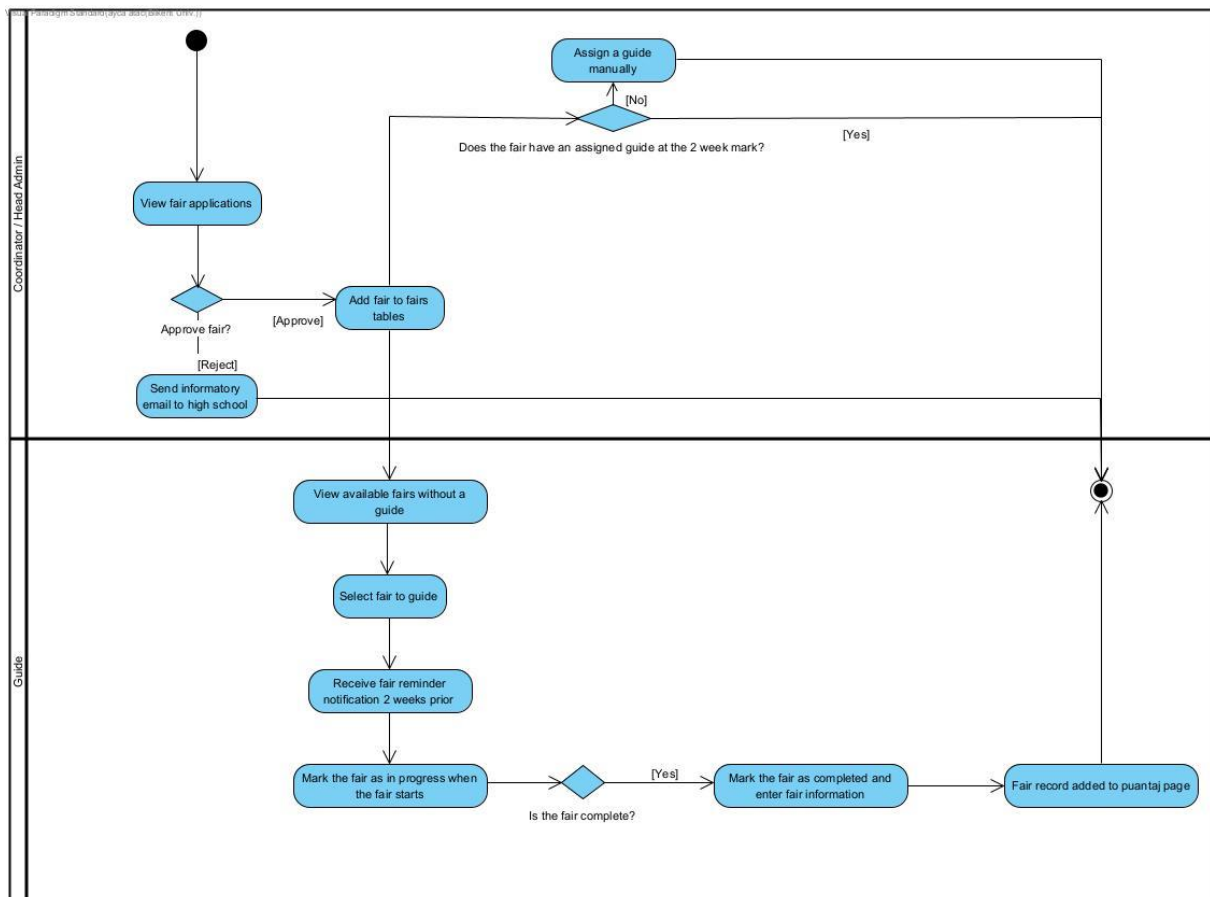


Figure 2: Fair Management System

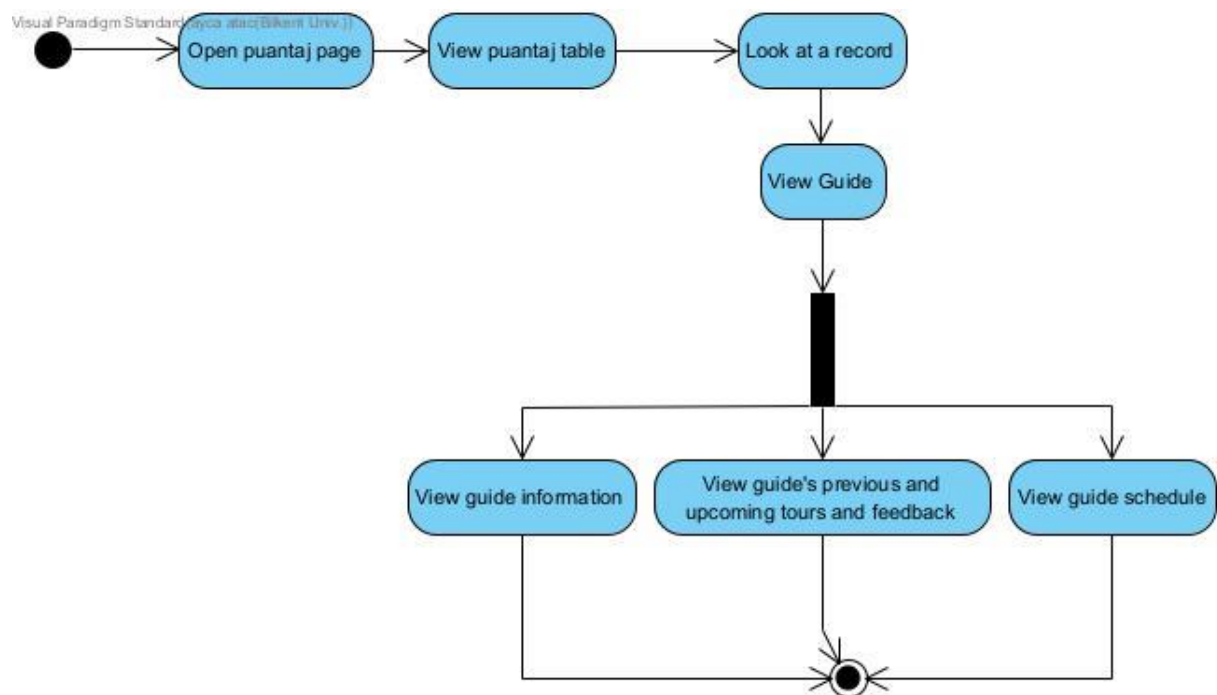


Figure 3: Point Records of Guides and Guide Information

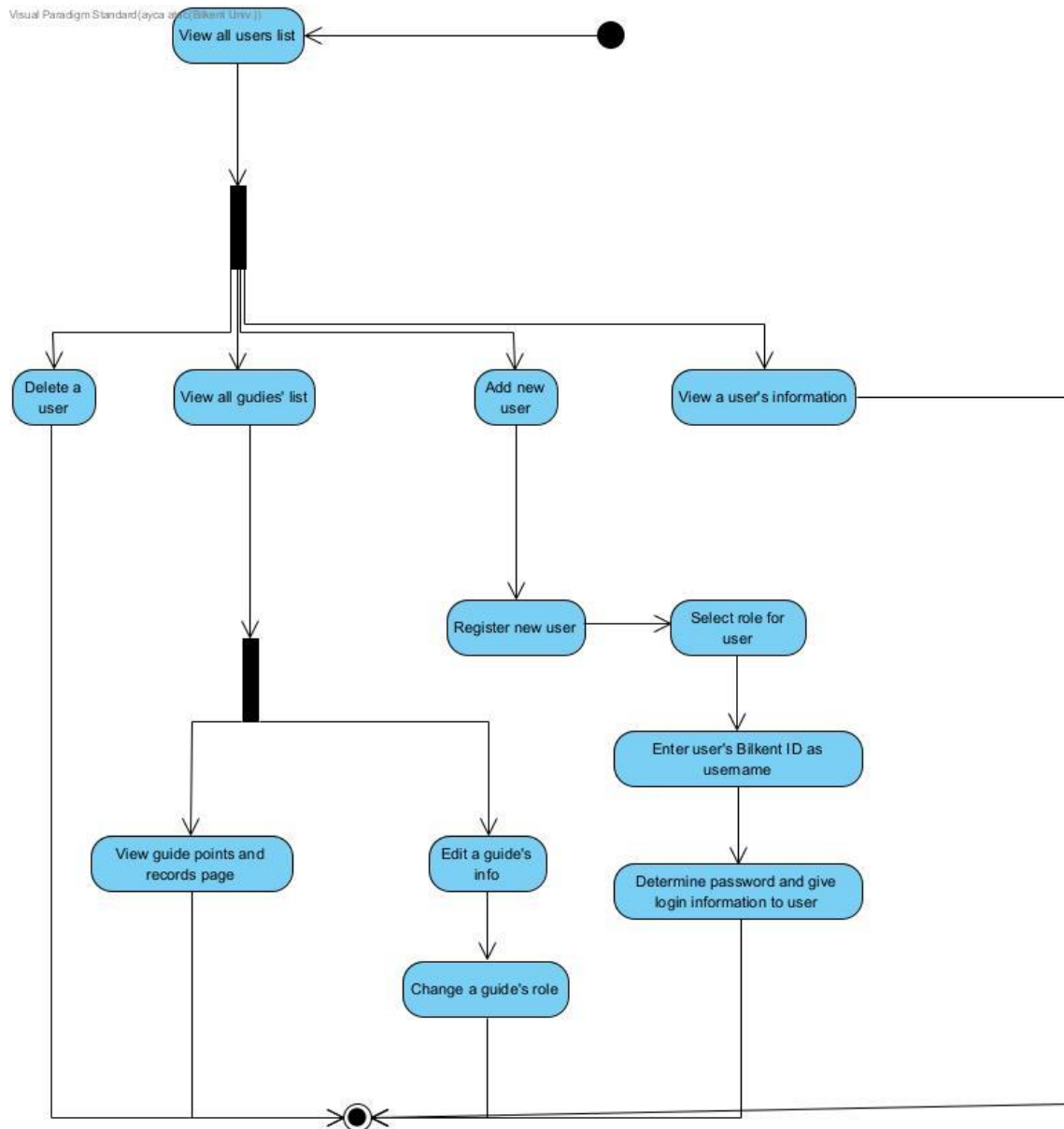


Figure 4: Manage Users

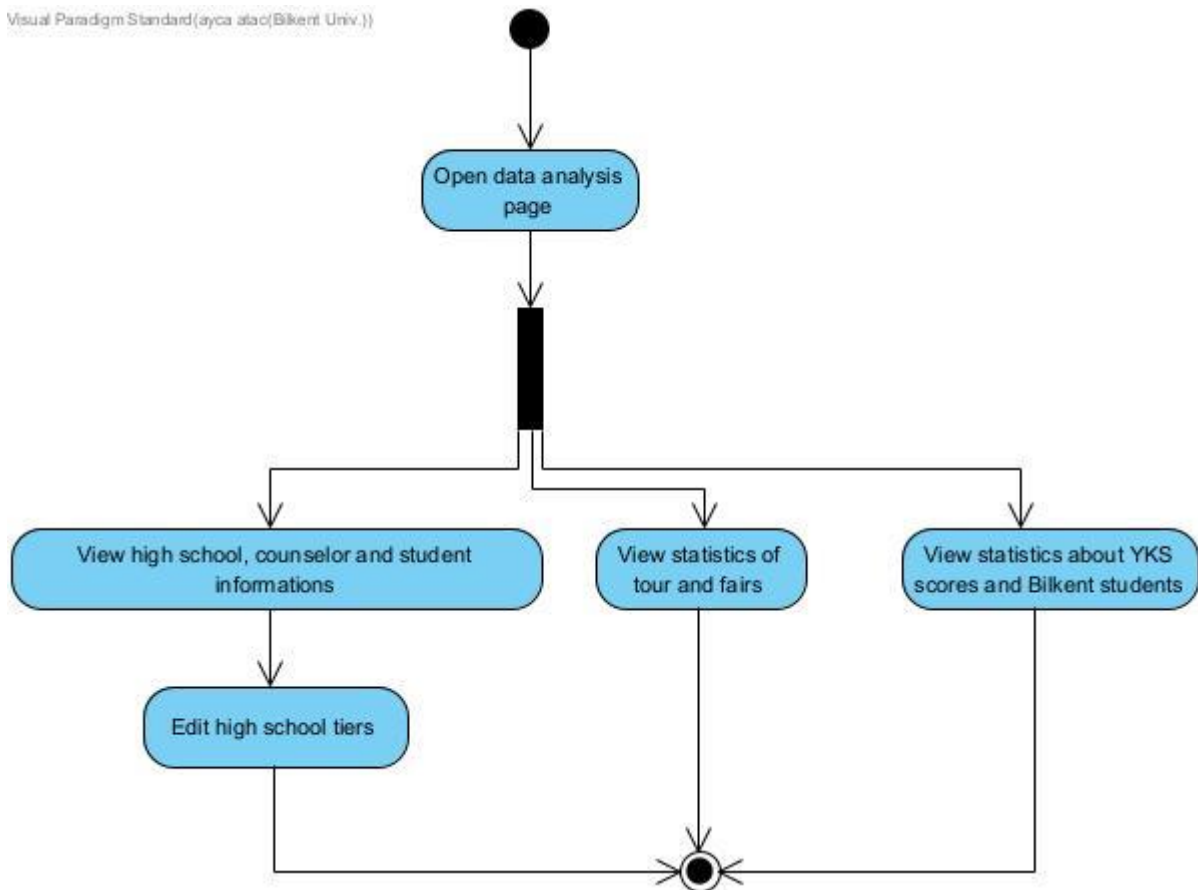


Figure 5: Data Analysis and Information Records

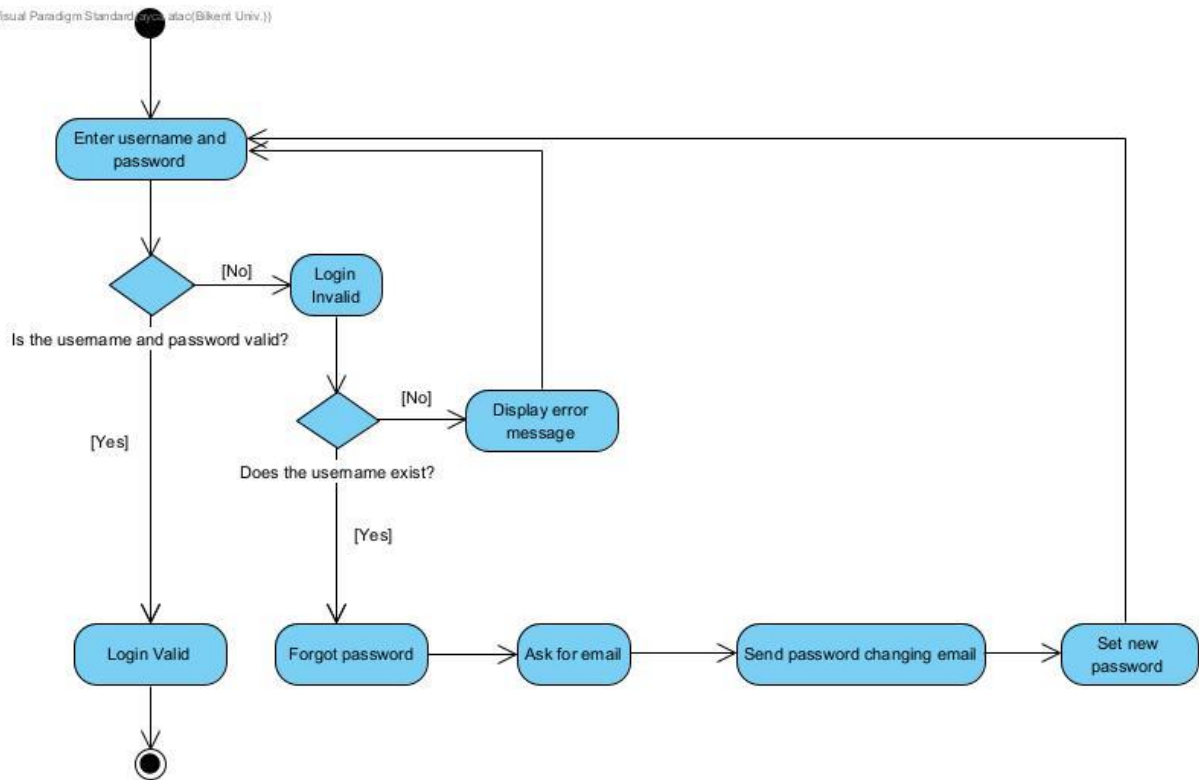


Figure 6: Login

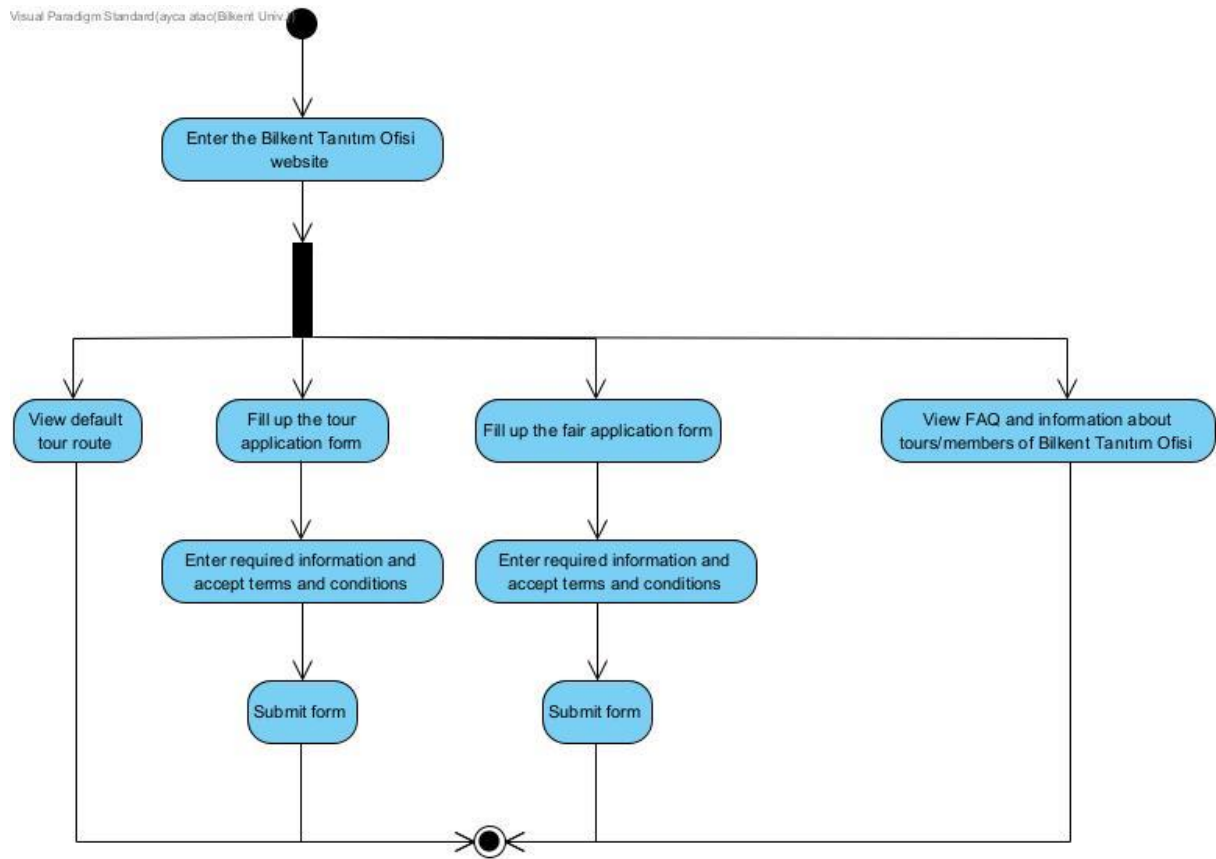


Figure 7: Entrance Page and Application Process for a Tour/Fair

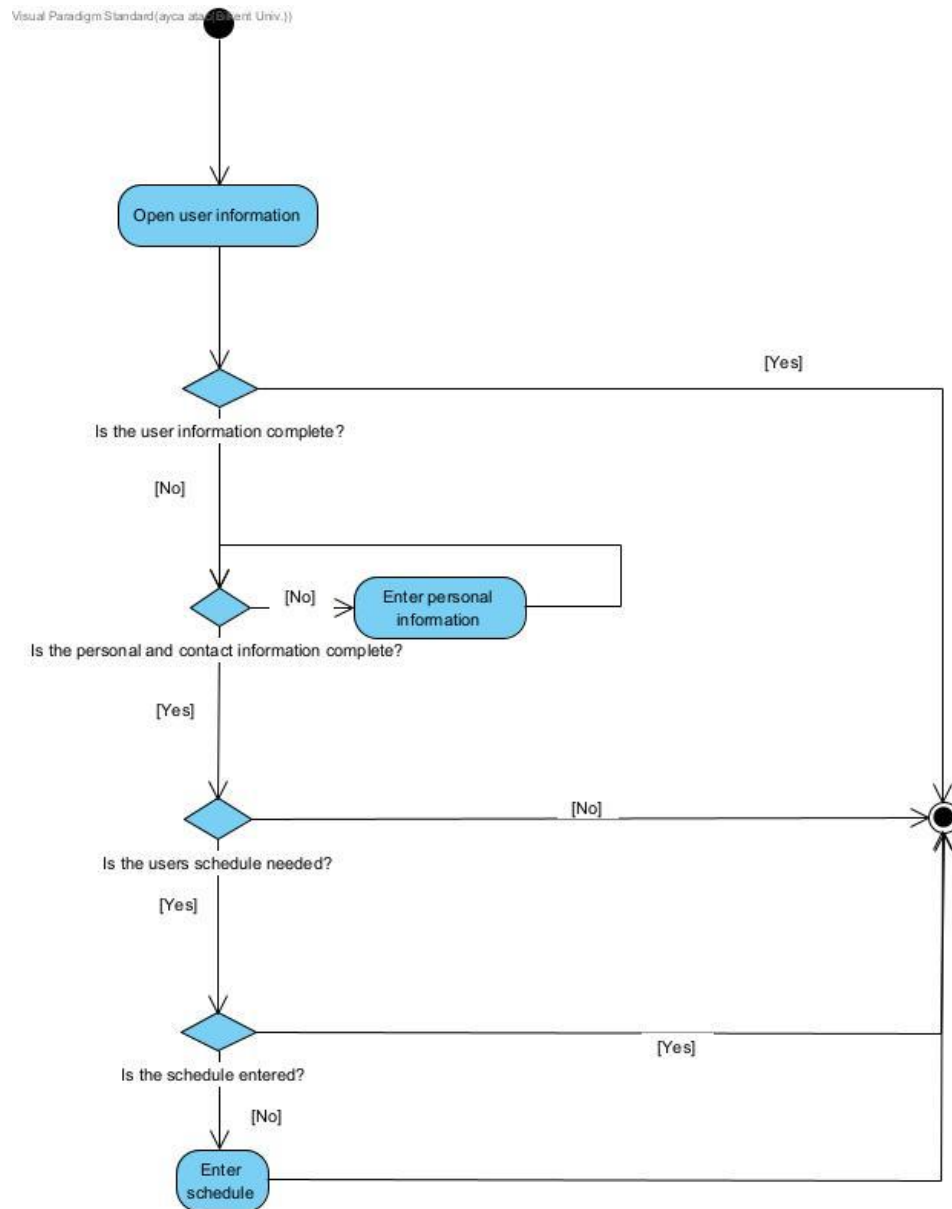


Figure 8: Complete Personal Information after Registration

3- Sequence Diagram

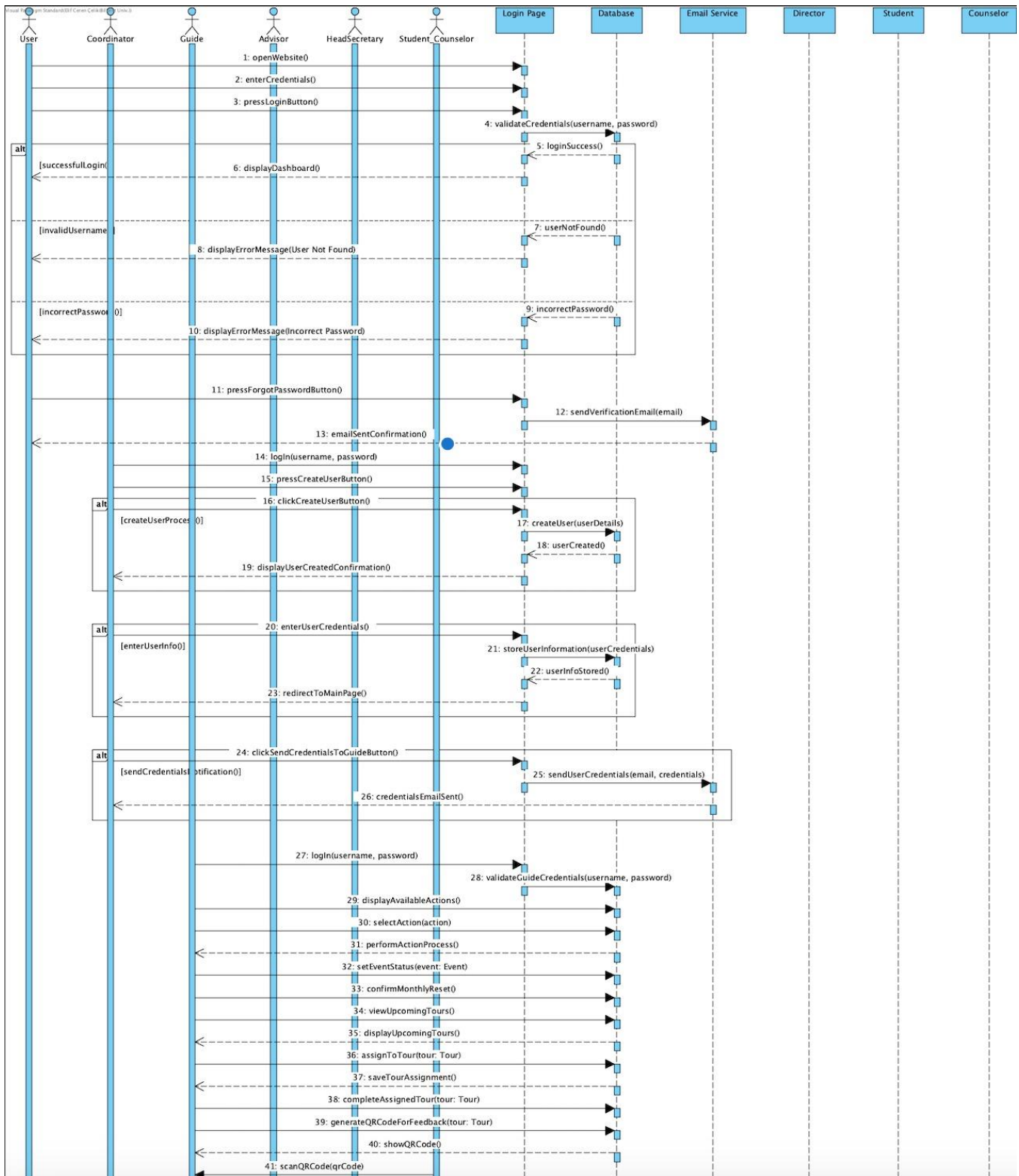


Figure 6: Overall Sequence Diagram Of The Application

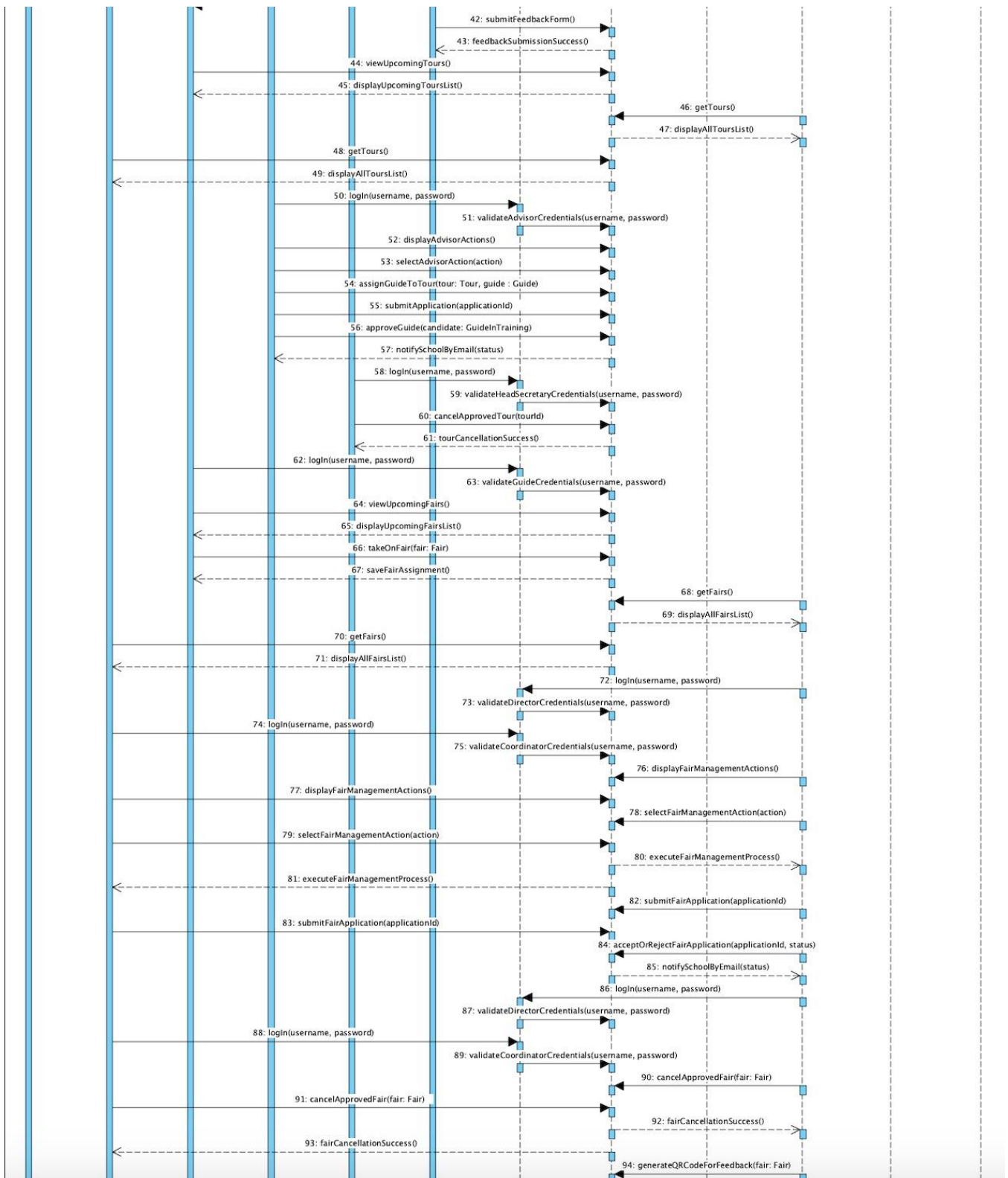


Figure 7: Sequence Diagram Continued



Figure 8: Sequence Diagram Continued

4- Class Diagram

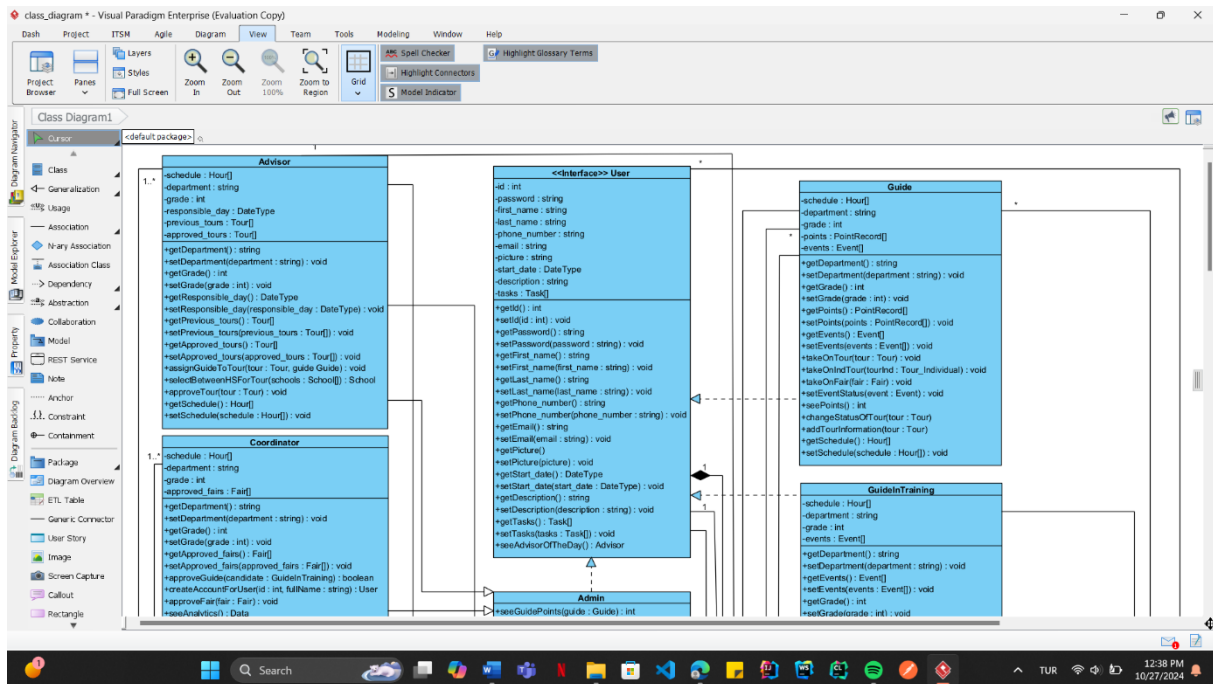


Figure 9: Advisor, Coordinator, Guide Classes and User Interface

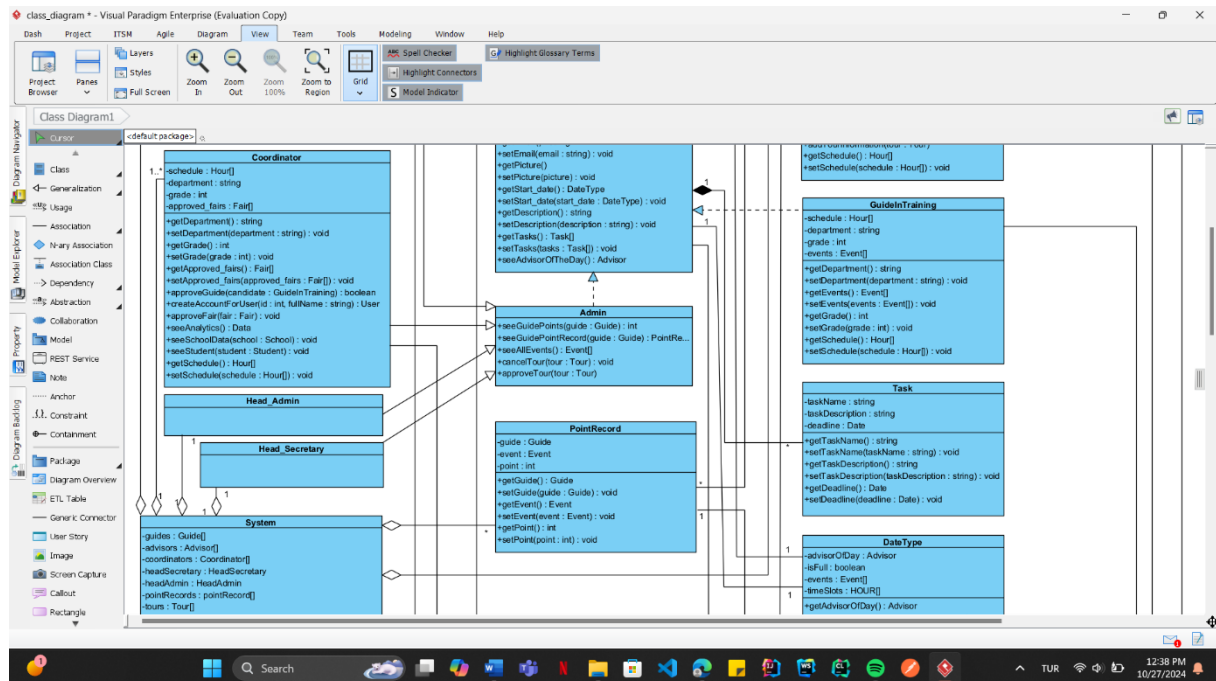


Figure 10: Head_Admin, Head_Secretary, System, Admin, PointRecord, GuideInTraining, and Task Classes

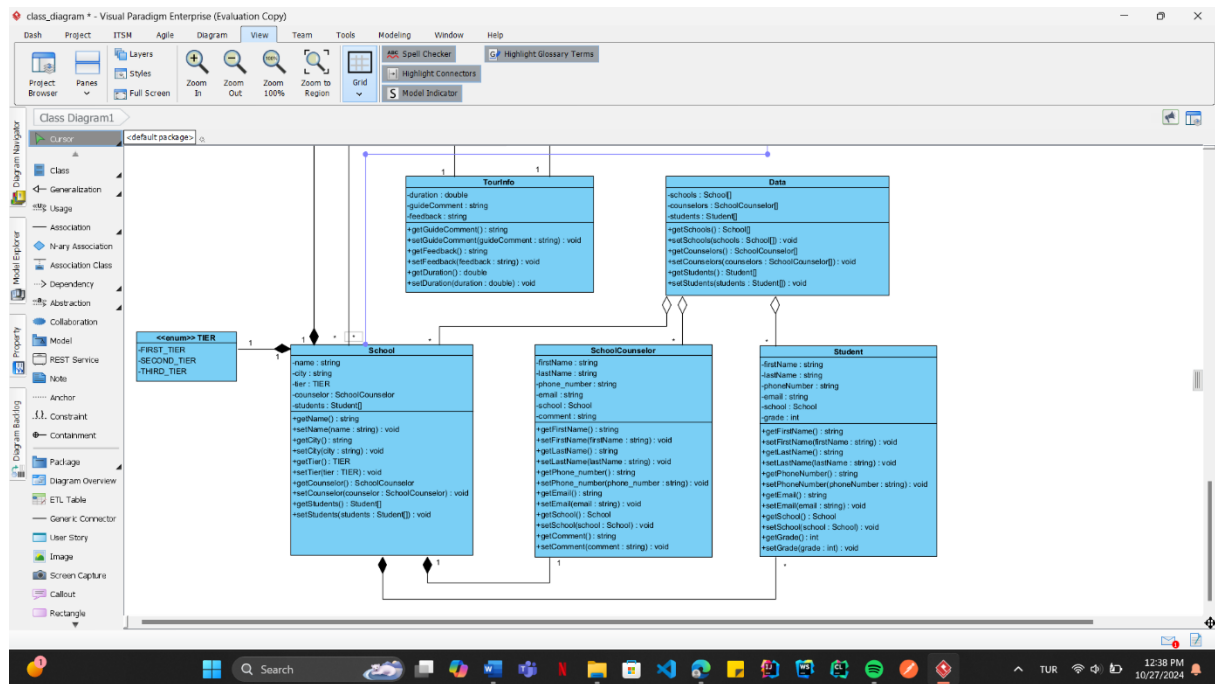
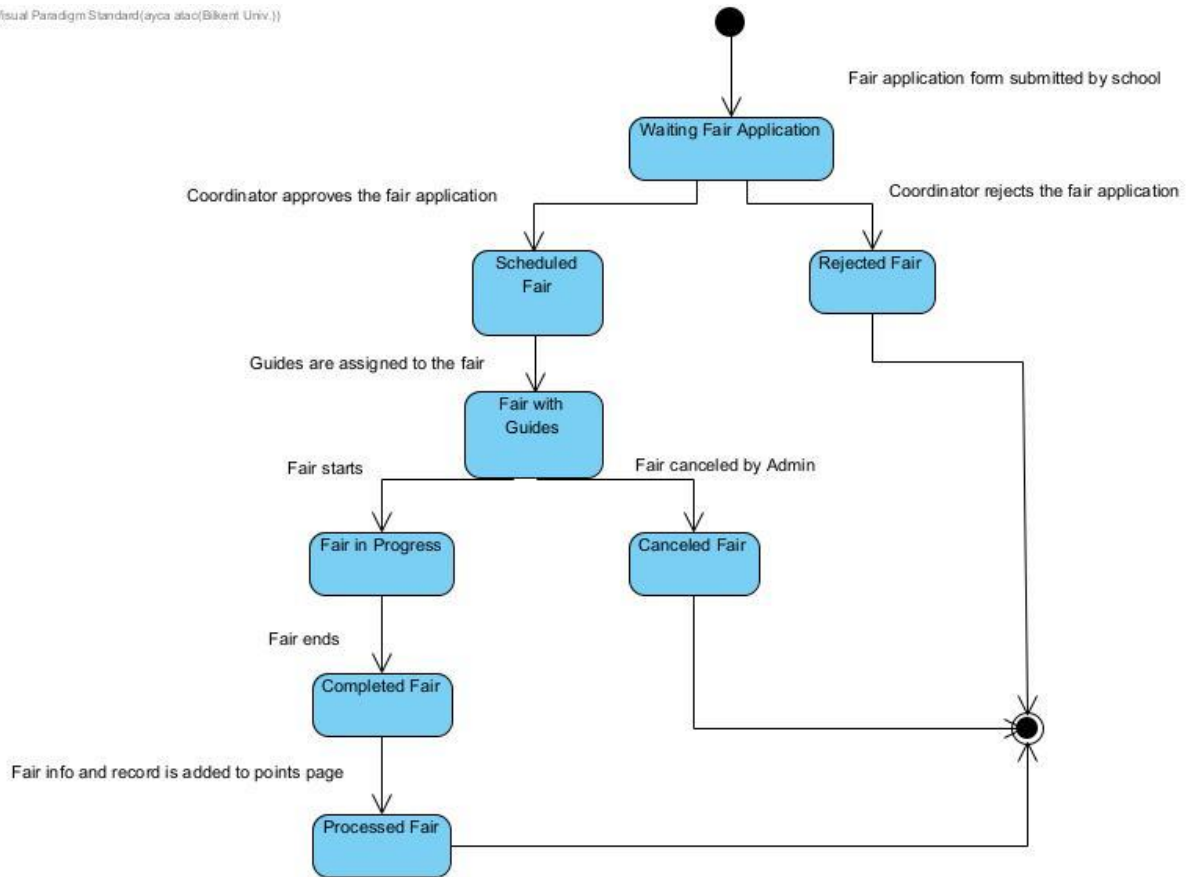


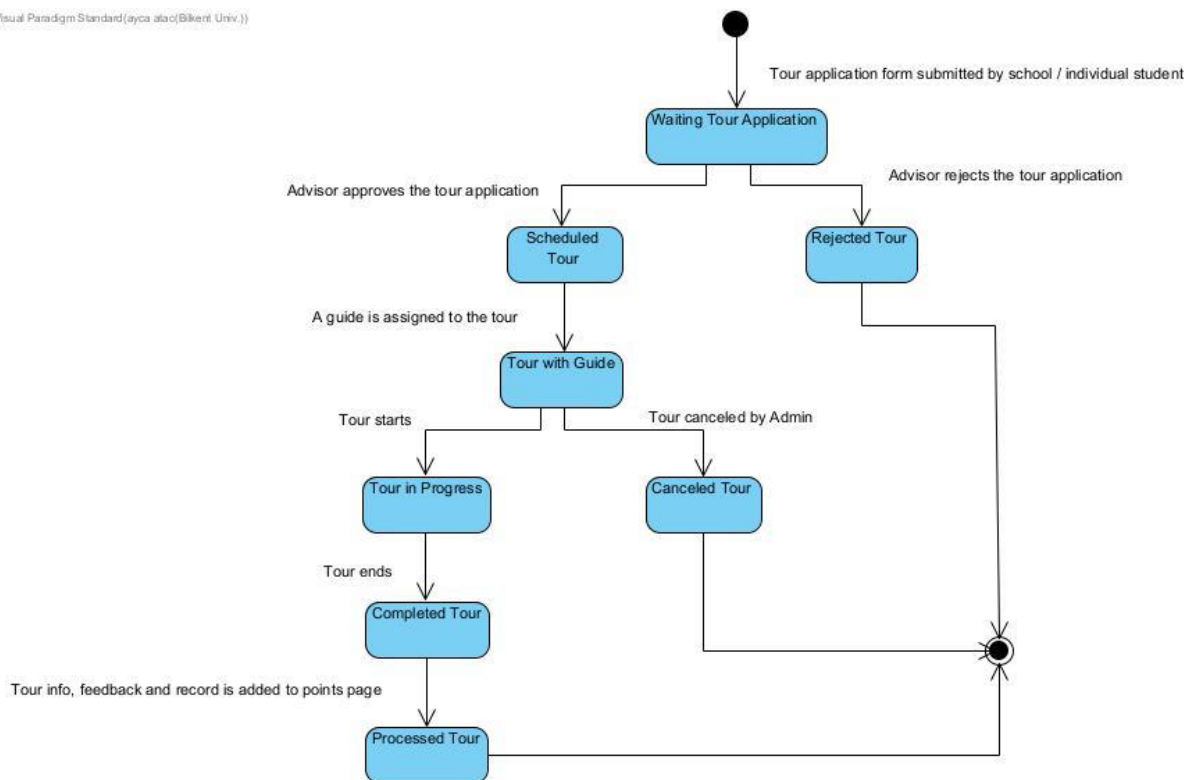
Figure 13: TIER enum; TourInfo, Data, School, SchoolCounselor, and School Classes

5- State Diagram

Visual Paradigm Standard (ayca at ac(Bilkent Univ.))

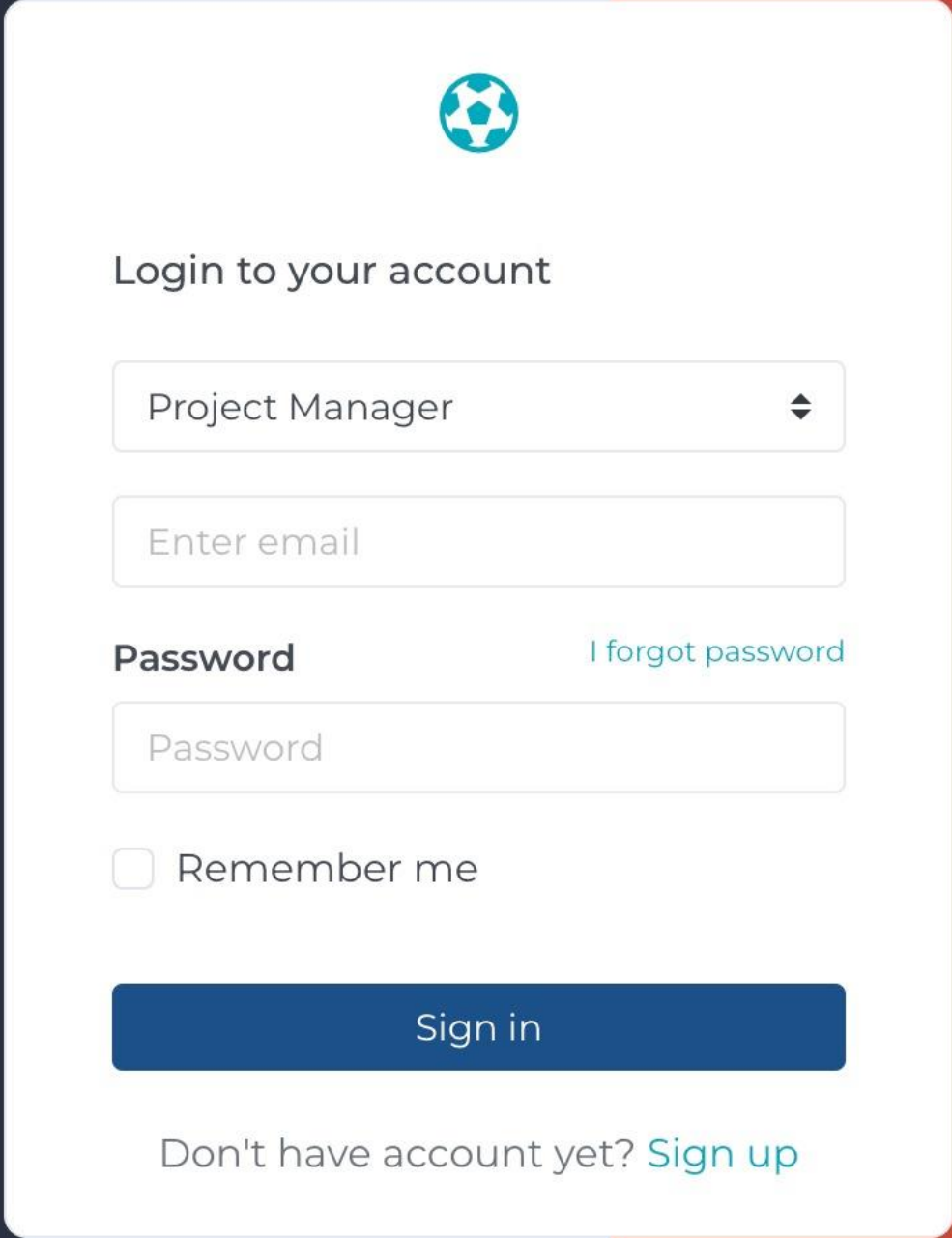


Visual Paradigm Standard (ayca at ac(Bilkent Univ.))



Figures 14 and 15: State Diagrams For Fair and Tour Processes Respectively

6- Mockups



A login form mockup centered on a background split into dark blue and red halves. The form is a white rounded rectangle containing a teal soccer ball icon, the title 'Login to your account', a role dropdown menu (currently showing 'Project Manager'), an email input field, a password section with a label, a 'Remember me' checkbox, a dark blue 'Sign in' button, and a link to 'Sign up' for new users.



Login to your account

Project Manager ▾

Enter email

Password [I forgot password](#)

Password

☐ Remember me

Sign in

Don't have account yet? [Sign up](#)



Forgot password

Enter your email address and your password will be reset and emailed to you.

Email address

Send me new password

Forget it, [Send me Back](#) to the Sign in screen.

Tour Summary

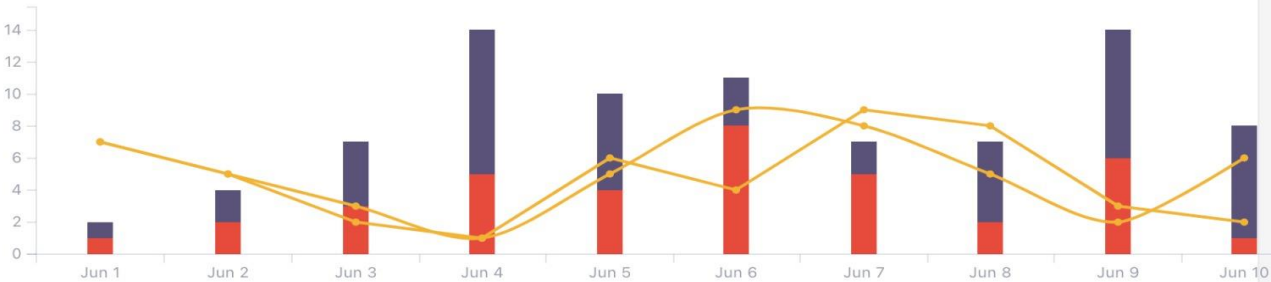
#	SCHOOL NAME	GUIDES	DATE	STUDENT NUMBER	STATUS
#AD1245	TED Ankara		17.05.2024	58	Pending
#DF1937	AAL	  	12.05.2024	120	Pending
#YU8585	OBL		28.04.2024	48	Done
#AD1245	İFL	   	20.04.2024	210	Done
#GH8596	Nesibe Aydın	  	12.04.2024	142	Done

Current School Status

^ [] x

as of Jun 2023 to 2024

1D 1W 1M 1Y



05

35%

NESIBE AYDIN

06

100%

BILKENT OBL

18

61%

AAL

11

87%

TED

Yesterday

Yesterday

Yesterday

Yesterday

School Satisfaction



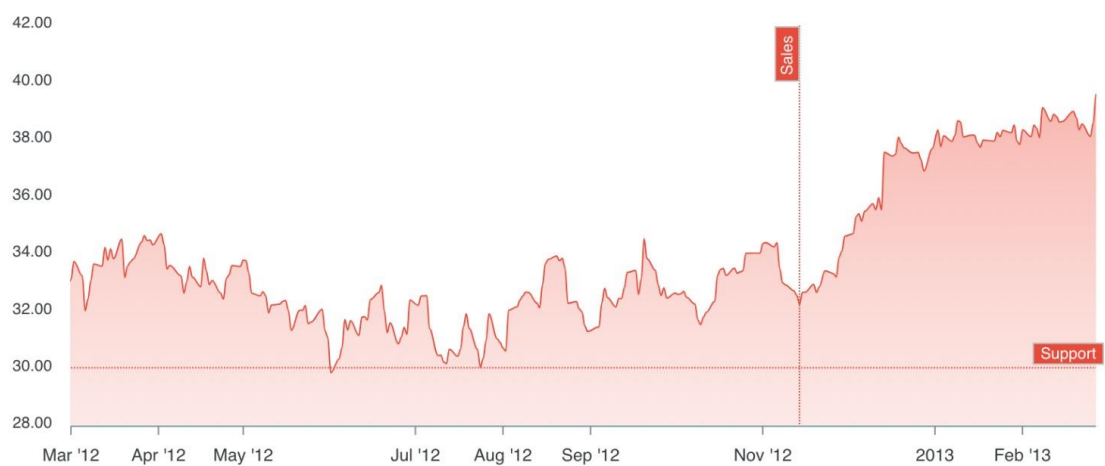
9.8 1.6% ↑
PERFORMANCE SCORE



● Excellent	3,007	50%
● Very Good	1,674	25%
● Good	125	6%
● Fair	98	5%
● Poor	512	10%

Guide Analytics

1M 6M 1Y **YTD** ALL



Guide



Ayse Dagli

ayse.dagli@bilkent.edu.tr

9.8 1.6% ↑

PERFORMANCE SCORE

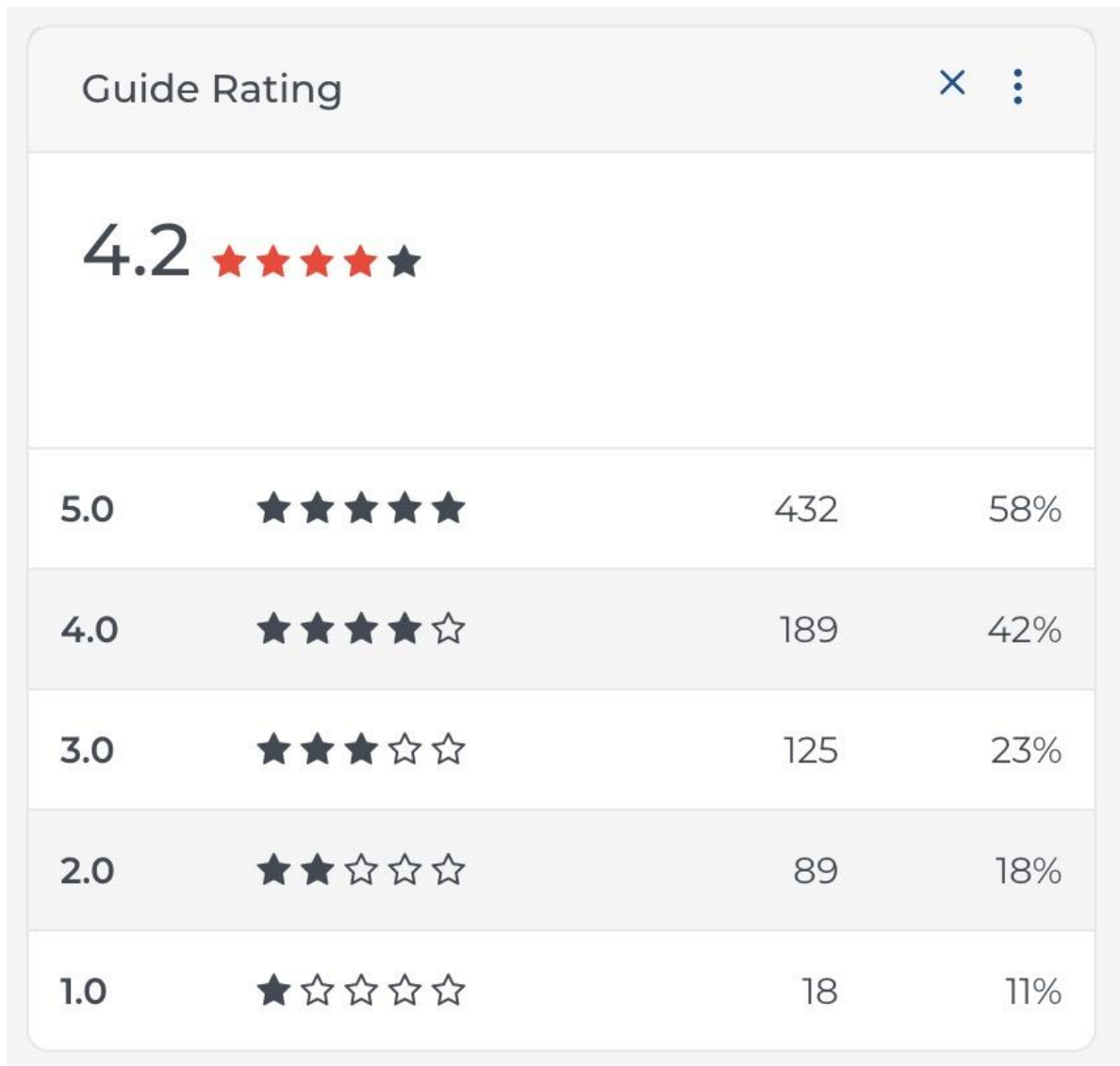
Activity

11:00am
Tour
TED Ankara

15.00
Meeting with Advisor

16:00pm
Meeting with Team
Hangouts





Guide Dashboard

Search for...

Welcome Ayse Dagli!

Pending Tours

2

More

Upcoming Tour

Ted Ankara

More

Pending all tours

7

View Fairs

12

More



Sara Hopkins



OCTOBER 2024

month

week

day



Sun	Mon	Tue	Wed	Thu	Fri	Sat
29	30	1 AAL Tour	2	3	4	5 OBL Tour
6	7	8	9 +2 more	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31	1	2
3	4	5	6	7	8	9